OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2
Examinations 2024-2025

Course Instance Code(s)	4BCT1, 4BSE1, 1MSME1,1MEEE1,1MECS1, 1CSD1, 1CSD2, 2SPE1, 1MAI1, 1MEES1
Exam(s)	B.Sc. Degree (Computer Science and Information Technology) B.E. Energy Systems Engineering, MSc in Mechanical Engineering, ME (Electrical and Computer Engineering), ME (Electrical and Electronic Engineering), M.Sc. in Computer Science (Data Analytics), M.Sc. in Computer Science (Artificial Intelligence)
Module Code(s)	CT561
Module(s) **Bonded**	Systems Modelling and Simulation No [☑] Yes [ ] List bonds
Paper No. **External Examiner(s)** **Internal Examiner(s)**	1 Dr. John Woodward Dr. Enda Howley *Prof. Jim Duggan

<u>Instructions</u>: Answer 3 questions out of 4. All questions will be marked equally.

Duration 2 hours

No. of Pages 5

Discipline(s) Computer Science

<u>Requirements:</u>

Remove Paper from exam venue, publish online	No [ ]	Yes[☑]
MCQ Answer sheet	No [☑]	Yes [ ]
Handout	No [☑]	Yes [ ]
Formulae & Log Tables**	No [☑]	Yes [ ]
Cambridge Tables 2nd Edition**	No [☑]	Yes [ ]
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [ ]	Yes [☑]
Other Materials	No [☑]	Yes [ ]
Graphic material in colour	No []	Yes [☑]

End of requirements

1. (a) Define a feedback loop, and describe the two different types of feedback.

    Summarise the advantages of adding feedback structures to a stock and flow model.

    [5]

    (b) Construct a stock and flow model from the following description of student workload. Clearly show the link polarities.

    There is a single Stock (**Assignment Backlog**) with two inflows (**Allocated** and **Rework**) and one outflow (**Completed**).

    1. **Assignment Backlog** is increased by the inflows **Allocated** and **Rework**, and reduced by the outflow **Completed**.
    2. As **Assignment Backlog** increases, so too does **Work Pressure**.
    3. An increase in **Work Pressure** leads to:
        - A decrease in **Time Per Assignment**
        - An increase in the **Weekly Study Hours**
        - An increase in **Amount of Time to Submit** (deadline extension by academic staff)
    4. An increase in **Time Per Assignment** leads to:
        - A reduction in flow **Rework**
        - A reduction in the flow **Completed**
    5. An increase in **Weekly Study Hours** leads to:
        - An increase in the flow **Completed**
        - An increase in **Fatigue**
    6. The remaining links are:
        - An increase in **Amount of Time to Submit** leads to a decrease in **Work Pressure**
        - An increase in **Fatigue** leads to a decrease in the flow **Completed**

    [12]

    (c) Based on the stock and flow model from part (b):

    - Show (and name) the three negative feedback loops that can be used for managing the assignment backlog as the work pressure increases. Confirm their loop polarity by tracing the effect of an increasing backlog through each loop.

    - Show (and name) the two positive feedback loops that are side-effects of a number of these policies. Confirm their loop polarity by tracing the effect of an increasing backlog through each loop

    [8]

2. (a) Describe, showing relevant examples, the following flows:

- Fractional Increase Rate
- Fractional Decrease Rate
- Adjustment to a Goal

[6]

(b) Explain how the *adjustment to a goal* structure can be extended to the stock management structure. When explaining the stock management structure describe: (1) the anchor and (2) the adjustment.

Show the delay structure that is used to construct the anchor in the stock management structure.

[7]

(c) Design a stock and flow model (with equations) that models staff recruitment in a University as a Stock Management Structure. Where appropriate, show the feedback loops in the model.

Here is the model information:

- The hire rate for staff is a function of an anchor and adjustment, and staff quits are based on a fractional decrease rate. Assume that the staff value is 500, which means it should start in equilibrium.

- The desired number of staff is based on the number of students and the desired student to staff ratio.

- Assume the desired student to staff ratio is 20

- Students are increased via a recruitment fraction, decreased via a graduation fraction, and are assumed to be at equilibrium (10000)

- Assume that the model starts in equilibrium and after 10 time units *the desired student to staff ratio drops from 20 to 10*.

Based on this:

1. Plot how you think the number of staff will change from 10 time units onwards (assume the simulation runs until time 30).

2. What auxiliary value in the model will have a big influence on how quickly the model resets to a steady state?

[12]

3. (a) Suppose we have a town with 10,000 (=N) individuals, of which 2% were infectious with measles, with R_0 = 12 and D=7 days.

- Calculate the force of infection λ.
- Calculate the Herd Immunity Threshold.

[5]

(b) Design a stock and flow model (and show equations and feedbacks) to simulate the spread of influenza.

Assume that:

- the value for R_0 = 2 (where R_0 is the average number of infections generated by an infectious person in a totally susceptible population).

- the average recovery delay = 2 days,

- the total population N = 10,000, with one of these initially infected.

The model should have the following features:

- Its core structure should be based on the *Susceptible – Infected – Recovered* model.

- It should allow for quarantine, though a quarantine fraction Q_F. A proportion of those infected will quarantine. People in quarantine will recover at the same rate as those who are infected.

- Assume Q_F = 0.08.

- Quarantine should be activated/deactivated through the use of a control flag (where a flag has a value of 0 or 1).

- Explain the force of infection equation.

[15]

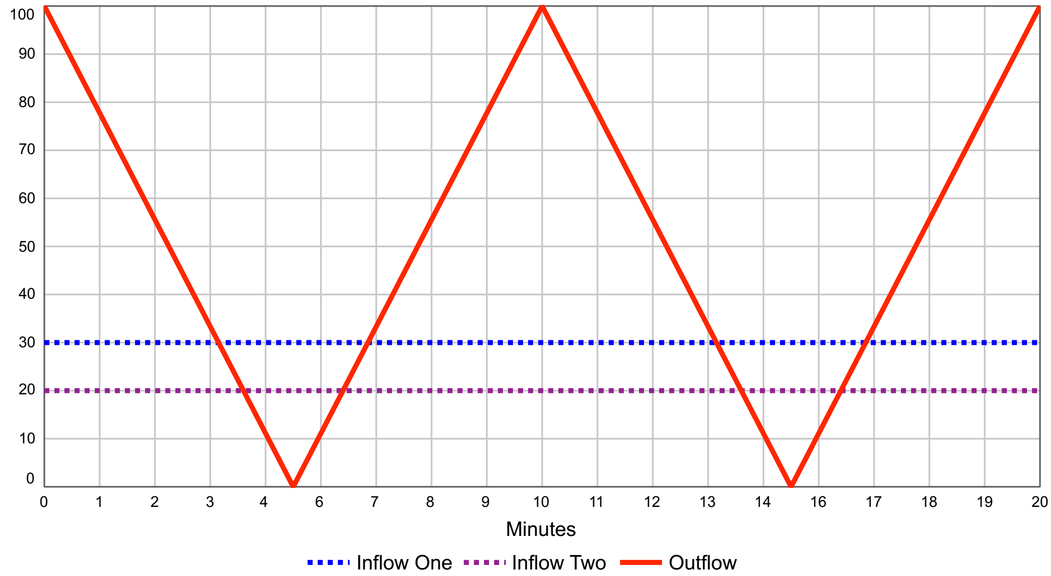
(c) Show how the model would need to be modified in order to represent different cohorts, for example, young and elderly in the population.

[5]

4. (a) Consider the three flow variables shown in the chart below.

- The outflow (in red) has a "sawtooth" shape beginning at 100 units/minute, dropping to 0 at time 5, and rising to 100 at time 10. The pattern repeats itself between time 10 and 20.
- There are two inflows, both constant. Inflow one (in blue) is 30 units/minute while inflow two (in magenta) is 20 units/minute

1. Draw a stock and flow model (including the net flow).
2. Calculate and show the net flow.
3. Use graphical integration to plot the stock over time, clearly explaining the different behaviour modes for each time interval. Assume the stock starts at 100 units.



[12]

(b) Describe Euler's method of integration, and state the underlying assumptions.

Under what conditions will Euler's method underestimate the true value of a stock? Support your answer by considering the numerical integration of the net flow over the time interval 0-5 from the model in part (a).

[5]

(c) Based on the flows shown in part (a), calculate the stock using Euler's equation where DT=10.

[8]

END